Chenxu Liu 刘宸旭

B.Sc. in Physics — Shandong University, Qingdao

202422201069@mail.sdu.edu · github.com/Phyliusir · Qingdao, China

EDUCATION

Shandong University, Qingdao

2024 – 2028

Institute of Frontier and Interdisciplinary Science · class of 2024, expected 2028

Physics for the foundations; almost all time outside it goes to engineering on both sides of large models and agents — read, reproduce, then change it on purpose.

RESEARCH INTERESTS

Large models: deployment and serving economics, post-training (SFT → preference → RL), and new attention architectures — Loop Attention.
Agents: agent construction, tool design, multi-agent collaboration, retrieval-augmented generation.

RESEARCH DIRECTIONS & WORK IN PROGRESS

- **Deployment (LM 01).** Cost structure of one model across serving stacks, batching and precision — cost per token, P99 latency and peak memory read together.
- **Post-training (LM 02).** A full SFT → preference → RL chain reproduced at small scale, then ablated stage by stage; focus on which gains survive on a smaller model.
- **Loop Attention (LM 03).** Reusing one attention block across depth — memory, wall-clock and training stability measured against a stacked baseline. Main line.
- **Agent construction (AG 01).** The smallest useful combination of loop, memory and planning; a taxonomy of tool-less agent failures as the starting object.
- **Tool design (AG 02).** Schemas, error surfaces and return shapes for a batch of real tasks, comparing failure distributions before and after.
- **Multi-agent collaboration (AG 03).** A strong single-agent baseline first; multi-agent setups have to beat it reliably before roles get split.
- **RAG (AG 04).** A citation-required QA set and a scoring script: an answer must land on a specific passage or it counts as wrong.

SELECTED PROJECTS

- **Loop Attention — depth-reuse comparison.** Experiment log and reproduction scripts; being organised.
- **Citation-required RAG evaluation set.** Question set plus scoring script; being organised.

TECHNICAL SKILLS

LANGUAGES	Python (primary), C++, CUDA, Bash, Git
MODEL SIDE	PyTorch training and fine-tuning, transformers and custom layers, serving and quantisation, LoRA and preference alignment
AGENT SIDE	Tools and function calling, multi-agent orchestration, vector retrieval and RAG, evaluation sets and scoring scripts
TOOLING	Linux and containers, distributed training basics, reproducible experiments

LANGUAGES

Chinese (native) · English (working proficiency).